

MIFE NORTHBOUND INTEGRATION

API SPECIFICATION

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INTRODUCTION

Northbound partners integrating to MIFE are expected to adhere to the API specification defined in this document. Adherence to the URL patterns and response codes defined herein is mandatory.

All APIs defined in this document are accessible via RESTful web services. A RESTful service utilizes HTTP commands POST, GET, PUT, and DELETE in order to perform an operation on a resource at the server. This resource is addressed by a URL; and what is returned by the server is a representation of that resource depending on its current state.

It is recommended that the communication between MIFE and the northbound partner should be via a secure channel (HTTPS).

Each API has parameters, which denote the API Version. This version is in alliance with the OneAPI specification on which these APIs are based on. Changing the API version may only be done with prior discussion and agreement with MIFE.

SECURITY

1. TOKEN API

Token API generates and renews user and application access tokens. Users need access tokens to invoke APIs subscribed under an application. Access tokens are passed in the HTTPS header when invoking APIs.

1.1. Generating Access Tokens with User Credentials (Password Grant Type)

You can obtain an access token by providing the resource owner's username and password as an authorization grant. It requires the base64 encoded string of the consumer-key:consumer-secret combination. You need the following before using the Token API to generate a token.

Prerequisites

- A valid user account in the API Store
- A valid consumer key and consumer secret pair
- These are to be created with help from MIFE Account Manager

1.2. Invoking the Token API to generate Tokens

Combine the consumer key and consumer secret keys in the format **consumer-key: consumer-secret** and encode the combined string using base64. Encoding to base64 can be done using the URL: <http://base64encode.org>.

Example

Consumer-Key:	wU62DjlyDBnq87GIBwplfqvmAbAa
Consumer-Secret:	ksdSdoefDDP7wpaElfqvmjDue

Concatenate the Consumer-Key & Consumer-Secret with a colon (:)
wU62DjlyDBnq87GIBwplfqvmAbAa:ksdSdoefDDP7wpaElfqvmjDue

And encode this string to base64, and you should get something like;
d1U2MkRqbHIEQm5xODdHbEJ3cGxmcXZtQWJBWTprc2RTZG9lZkREUDd3cGFFbGZxdm1qRHVl

Payload

grant_type=password&username=<username>&password=<password>&scope=PRODUCTION

Replace the<username> and <password> values as appropriate.

Headers

Authorization: Basic <base64 encoded string>, Content-Type: application/x-www-form-urlencoded

Replace the <base64 encoded string> as appropriate.

For example, use the following cURL command to access the Token API. It generates two tokens as an access token and a refresh token. You can use the refresh token at the time a token is renewed.

```
curl -k -d "grant_type=password&username=<username>&password=<password>&scope=PRODUCTION" -H "Authorization: Basic <authorization code>, Content-Type: application/x-www-form-urlencoded" "https://apigate.robi.com.bd/token"
```

1.3. Renewing access Token

After an access token is generated, you are required to refresh the old token every 3300 seconds (55 mins). This can be done by issuing the same REST call to the Token API through a REST client such as the WSO2 REST Client or Curl. It is good practice to build the token refresh logic into your application to ensure automatic renewal of token.

```
curl -k -d  
"grant_type=password&username=<username>&password=<password>&scope  
=PRODUCTION" -H "Authorization: Basic <authorization code>," Content-Type:  
application/x-www-form-urlencoded" "https://apigate.robi.com.bd/token"
```

Running this same command will return the exact same Token if it is run within 1 hour and the Token has not expired yet. However, it will return a fresh Token if it is run after the current Token's expiry.

1.4. Revoking access Token

After issuing an access token, a user or an admin can revoke it in case of theft or a security violation. You can do this by calling Revoke API using a utility like cURL. The Revoke API's endpoint URL is <Revoke Token URL>.

Parameters required to invoke this API are as follows:

- The token to be revoked
- Consumer key and consumer secret key. Must be encoded using Base64 algorithm

```
curl -k -d "token=<ACCESS_TOKEN_TO_BE_REVOKED>" -H "Authorization: Basic  
<authorization code>" <Revoke Token URL>
```

API SPECIFICATION

APIs discussed under this section are RESTful APIs with a JSON payload where applicable.

2. SEND SMS

HTTP Method	POST
Operation	Used to send an SMS from an application (to one or more mobile terminals).

2.1. Request

```
curl --location
'https://apigate.robi.com.bd/Ext/smsmessaging/v1/outbound/tel%3A%2B880
1833183***/requests' \
--header 'Authorization: Bearer <Token>' \
--header 'Content-Type: application/json' \
--data '{
  "outboundSMSMessageRequest": {
    "address": [
      "tel:+8801833183***"
    ],
    "senderAddress": "tel:+212839",
    "outboundSMSTextMessage": {
      "message": "Hi, it is test"
    },
    "clientCorrelator": "123457:AIN12345611493",
    "receiptRequest": {
      "notifyURL": "http://127.0.0.1/rest_test.php",
      "callbackData": "some-data-useful-to-the-requester",
      "senderName": "2128***"
    }
  }
}'
```

2.2. Request Parameters

Parameter name	Description	Usage
address	At least one address will be provided; in this case the recipients MSISDN including the 'tel:' protocol identifier and the country code preceded by '+'. i.e., tel:+8801833xxxxxx.	Mandatory, one or more
senderAddress	(String) is the address to whom a responding SMS may be sent	Mandatory
message	(String) will be provided within the outboundSMSTextMessage element. Messages over 160 characters will be counted as two or more messages.	Mandatory
clientCorrelator	(String) uniquely identifies this create SMS request.	Mandatory

senderName	(String) is the URL-escaped name of the sender/masking to appear on the terminal.	Optional
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2.3. Response

```
{
  "outboundSMSMessageRequest": {
    "address": [
      "tel::+8801833183****"
    ],
    "deliveryInfoList": {
      "deliveryInfo": [
        {
          "address": "tel:+8801833183****",
          "deliveryStatus": "MessageWaiting"
        }
      ]
    },
    "resourceURL":
      "https://apigate.robi.com.bd/smsmessaging/v1/outbound/tel:2128***/requests/AIN12345611493/deliveryInfos"
  },
  "senderAddress": "tel:2128***",
  "outboundSMSTextMessage": {
    "message": "Hi, it is test"
  },
  "clientCorrelator": "123457:AIN12345611493",
  "receiptRequest": {
    "notifyURL": "http://127.0.0.1/rest_test.php",
    "callbackData": "some-data-useful-to-the-requester"
  },
  "senderName": "",
  "resourceURL":
    "https://apigate.robi.com.bd/smsmessaging/v1/outbound/tel:2128399/requests/AIN12345611493"
}
```

3. RESPONSE CODES

HTTP response codes are used to indicate:

- 200 – Success!
- 400 – Bad request; check the error message for details
- 401 – Authentication failure, check your authentication details
- 403 – Forbidden; please provide authentication credentials
- 404 – Not found: mistake in the host or path of the service URI
- 405 – Method not allowed: for example you mistakenly used a HTTP GET instead of a POST
- 500 – The server encountered an unexpected condition. This could be wrong authentication details or limited user permission
- 503 – Server busy and service unavailable. Please retry the request.